

Reading an Enclosure Rating: IP & NEMA

Two systems say the same thing two ways. International spec sheets print IP (IEC 60529); US sheets print NEMA (NEMA 250). Decode both - and know which direction the translation runs.

PART 1

Decode the IP code

IP

PREFIX

Ingress Protection

Defined by IEC 60529. Each digit = a fixed lab test, which is why the code is trustworthy and "waterproof" is not.

6

1ST DIGIT · SOLIDS

6 - Dust-tight

Object gate 50 → 12.5 → 2.5 → 1 mm, then a dust chamber at 5–6.

7

2ND DIGIT · WATER

7 - 1 m immersion, 30 min

A separate water test, 0–9K. Independent of the solids digit.

1st digit - solids & dust	2nd digit - water
0 No protection	0 None
1 ≥50 mm · back of the hand	1 Vertical drip
2 ≥12.5 mm · finger	2 Drip, tilted 15°
3 ≥2.5 mm · tools, thick wire	3 Spray to 60° from vertical
4 ≥1.0 mm · fine wire, screws	4 Splash from any direction
5 Dust-protected · limited ingress, no interference	5 Jets · 6.3 mm nozzle
6 Dust-tight · no ingress at all	6 Powerful jets · 12.5 mm nozzle
	7 Immersion · 1 m, 30 min
	8 Immersion >1 m · depth/time set by maker
	9K High-pressure hot jets · 80 °C

Common ratings a WLAN pro reads	
CODE	PLAIN MEANING & TYPICAL EXAMPLE
IP54	Dust-protected + splash. Covered-soffit / IoT sensor.
IP65	Dust-tight + low-pressure jets. Sheltered outdoor.
IP66	Dust-tight + powerful jets. Common outdoor AP / panel antenna.
IP67	Dust-tight + 30 min at 1 m. The mainstream outdoor / rugged AP bar.
IP68	Dust-tight + deeper immersion, <i>maker-defined</i> depth/time.
IP69K	High-pressure hot wash. Car wash / food-plant washdown.
IP6X	Dust-tight; water not rated (X = untested, not "fails").
READ X RIGHT IPX7 means the water rating is 7 and solids were never tested - it is not IP07. X = "no data," never "zero."	

PART 2

IP ↔ NEMA - a one-way bridge

IP

IEC 60529 · international

Two digits, each a fixed ingress test: solids/dust, then water. Nothing else - no corrosion, icing, or gasket-aging check.

1st - solids & dust (0–6)

2nd - water (0–9K)

NEMA → IP
maps to a minimum IP

IP → NEMA
does NOT imply a NEMA type

NEMA

NEMA 250 · US (ANSI/NEMA 250-2020)

Also tests corrosion, external icing, gasket aging, oil, and construction - a broader durability rating, not just ingress.

3R - rain / sleet / snow [Wi-Fi]

4 - + hose-down, watertight [Wi-Fi]

4X - Type 4 + corrosion (coastal) [Wi-Fi]

6 / 6P - + submersion [Wi-Fi]

NEMA 3R → IP14 / IP24 varies

NEMA 4 / 4X → IP66

NEMA 6P → IP67 / IP68 varies

Why the arrow only runs one way: NEMA tests corrosion, icing, gasket aging, and oil - things IP never checks. A NEMA 4X box exceeds IP66 for ingress, but an IP66 box is **not** automatically NEMA 4X (it was never corrosion-tested). Map NEMA → IP as a **minimum**; never the reverse.

MYTH A bigger number is **not** always more protection. IP67 (immersion) does not guarantee IP66 (jets) - the water ladder is not a clean superset past 6. Gear that must survive both is dual-marked (IP66/IP67). Match the digit to the actual hazard.

Reference only. NEMA→IP equivalents are approximate minimums, not a certified conversion (NEMA states the types "cannot be exactly converted"); 3R and 6P vary by source. Confirm the specific listing and the enclosure's temperature and corrosion specs against the vendor data sheet.

WLAN PROS · ENCLOSURE RATINGS